

Name: _____ Student ID: _____

EGS3012S (2024) Test 2A

Total: 23 (MCQ)+12 (other) = 35 marks (~70s/question)

MCQ: Correct answer = +1, Incorrect = -0.25, Blank = 0 (Negative marking)

#	Question	Answer
1	<u>An anticyclonic gradient wind speed is:</u> (A) supergeostrophic (B) equal in speed to the geostrophic wind (C) subgeostrophic (D) centripetal	
2	<u>The geostrophic approximation closely approximates real winds:</u> (A) Near the Earth's surface, where friction is high (B) In the upper troposphere, where friction is low (C) At the Equator, where the Coriolis parameter is 0 (D) Around mountains (E) Over forests	
3	<u>Surface Southern Hemisphere cyclones are characterised by flow that is:</u> (A) Anti-clockwise, downward and outward (B) Anti-clockwise, upward and inward (C) Clockwise, upward and outward (D) Clockwise, downward and outward (E) Clockwise, upward and inward	
4	<u>If the wind direction is 90°, the wind is coming from the:</u> (A) West (B) North (C) East (D) South-West	
5	<u>What is the difference in angle of elevation of the sun at solar noon between the summer and winter solstices at Durban (30°S, 30°E)?:</u> (A) 0° (B) 23.5° (C) 30° (D) 47° (E) 60°	
6	<u>For down-slope winds the dominant force is</u> (A) Buoyancy (B) Coriolis force (C) Centrifugal force (D) Gravity	
7	<u>Which of the following types of idealised wind accounts for the Coriolis effect?</u> (A) Sea-breeze (B) Valley breeze (C) Gradient wind (D) Cyclostrophic wind	

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8	<u>Which of the following is considered an adiabatic process?</u> (A) Conduction heats air immediately above the Earth's surface (B) An air parcel rising, expanding and cooling without mixing with its environment (C) Rain evaporates while falling from the clouds before it reaches the ground (D) Snow cover is sublimated into a dry air parcel moving over it	
9	<u>Where are most of the world's hot deserts situated relative to the tri-cellular arrangement?</u> (A) At the Inter-Tropical Convergence Zone (ITCZ) (B) At the poleward edge of the Hadley Cells (C) At the centre of the Ferrel Cells (D) At the Equatorward edge of the Polar Cells	
10	<u>Which of the following could strengthen a sea-breeze circulation during the day?</u> (A) Increased soil moisture (B) Decrease in sea surface temperatures (C) Strong background flow from the land to the sea (D) Increased uniform cloud cover	
11	<u>For questions 11–14, consider the following 4 air parcels (all starting from the surface at sea level with surface pressure $P=1000\text{hPa}$), characterised as follows (where CAPE is convective available potential energy; CIN is convective inhibition; LCL is the level of free convection; and EL is the equilibrium level):</u> A: $\text{CAPE}=2000\text{J}\cdot\text{kg}^{-1}$; $\text{CIN}=1500\text{J}\cdot\text{kg}^{-1}$; $\text{LCL}=830\text{hPa}$; $\text{EL}=130\text{hPa}$ B: $\text{CAPE}=1000\text{J}\cdot\text{kg}^{-1}$; $\text{CIN}=400\text{J}\cdot\text{kg}^{-1}$; $\text{LCL}=920\text{hPa}$; $\text{EL}=440\text{hPa}$ C: $\text{CAPE}=100\text{J}\cdot\text{kg}^{-1}$; $\text{CIN}=700\text{J}\cdot\text{kg}^{-1}$; $\text{LCL}=970\text{hPa}$; $\text{EL}=680\text{hPa}$ D: $\text{CAPE}=100\text{J}\cdot\text{kg}^{-1}$; $\text{CIN}=1200\text{J}\cdot\text{kg}^{-1}$; $\text{LCL}=670\text{hPa}$; $\text{EL}=510\text{hPa}$ <u>In which of the air parcels is a thunderstorm most likely to occur?</u>	
12	<u>In which of the air parcels would updrafts produced in a thunderstorm likely be most vigorous?</u>	
13	<u>For which of the air parcels would the cloud top height (in m) likely be greatest?</u>	
14	<u>For which of the air parcels is the cloud base likely to be closest to the ground?</u>	
15	<u>Which of the following atmospheric layers will always be stable?</u> (A) An inversion layer (B) A layer where environmental potential temperature decreases with height (C) A layer in which environmental equivalent potential temperature decreases with height (D) The day time atmospheric boundary layer	

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16	<u>If the near-surface air temperature at 1000hPa equals 3°C and the dew point temperature 1°C, the lifting condensation level height is approximately</u> (A) At the surface (B) At the tropopause (C) 20m (D) 200m (E) 2000m	
17	<u>What prevents wind from following the direction of the horizontal pressure gradient force?</u> (A) Friction with the ground (B) Gravity (C) Centrifugal acceleration (D) The Coriolis effect	
18	<u>The marine atmospheric boundary layer during the day is:</u> (A) Thicker than the boundary layer over land (B) Shallower than the boundary layer over land due to a lack of moisture availability (C) Shallower than the boundary layer over land but the diurnal mean thickness of the marine boundary layer can be greater than that of the boundary layer over land (D) Shallower than the boundary layer over land and the diurnal mean thickness of the marine boundary layer is always less than that of the boundary layer over land	
19	<u>When upper-level divergence of air above a surface low pressure area is stronger than the convergence of surface air, the surface pressure will:</u> (A) Increase (B) Stay the same (C) Decrease (D) Become negative	
20	<u>Which of the following is not correct about thermal wind?</u> (A) Thermal wind is parallel to geostrophic wind in a baroclinic atmosphere (B) Westerlies strengthen with height as thermal wind is additive to the geostrophic wind (C) Easterlies weaken with height as they are opposed by thermal wind (D) Cold cored cyclones intensify with height and are deep	
21	<u>During the day time the atmospheric boundary layer has an entrainment zone above which entrains:</u> (A) Free atmosphere air into the mixed layer (B) Free atmosphere air into the surface layer (C) Surface layer air into the free atmosphere (D) Mixed layer air into the surface layer	
22	<u>In a conditionally unstable atmospheric column with a lifting condensation level height of 3000m, for which of the following would the pressure value be lowest?</u> (A) The lifting condensation level (B) The level of free convection (C) The equilibrium level (D) The boundary layer capping inversion	

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23 Which of the following is **not** true:

- (A) Cloud development frequently occurs at the sea breeze front
- (B) Sea breezes form because land heats up and cools faster than water
- (C) Sea breeze circulation is a thermally induced seasonal reversal of winds
- (D) Sea breeze circulation is formed because of a pressure difference between land and sea

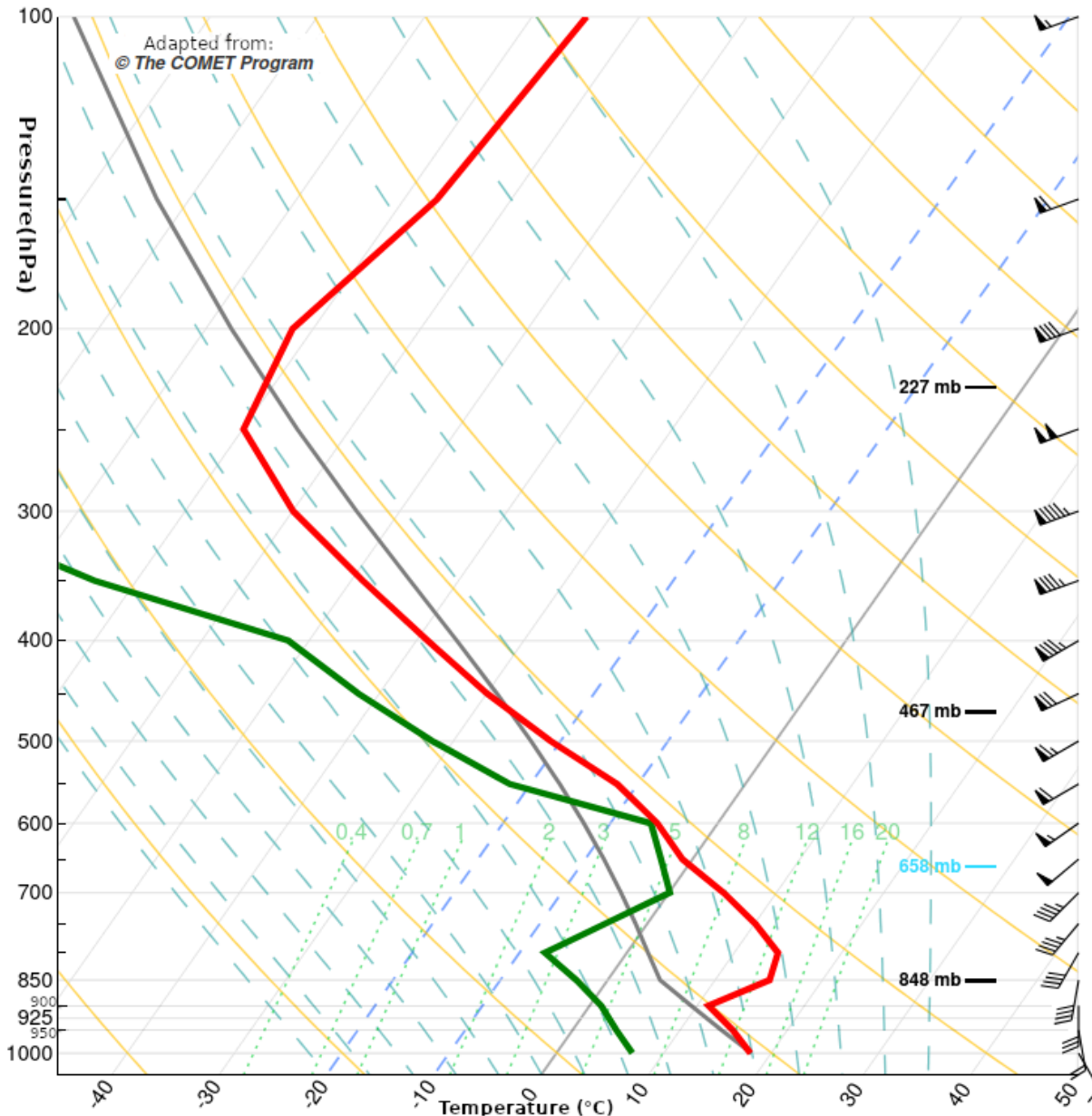


Illustration for Question (B) below. The green solid line is the T_d (dew point temperature) profile, the red solid line (to the right of the green one) is the environmental temperature profile (T_e), the grey solid line is the parcel temperature (T_a) profile for a parcel lifted adiabatically from the surface. Significant pressure values, including the lifting condensation level, level of free convection and equilibrium level, are labelled along the right-hand side. Note that 1mb = 1hPa. The light blue mark at 658mb is the freezing level. Grey solid horizontal lines are isobars. Black marks on the left vertical axis are spaced at 50hPa. Grey solid lines sloping upward to the right are isotherms (the -20 and -10°C isotherms are blue dashed lines and the 0°C isotherm is thicker than the others). Saturation mixing ratio (X_s) isohumes are dotted lighter green lines, labelled at the 600hPa level in units of g/kg. Solid orange lines are dry adiabats, the dashed dark green lines are saturated adiabats.

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Calculation question:

- (A) Between Cape Town and Caledon, 100km to the east, the 500hPa geopotential height decreases from 5200 to 5180m, whereas along the meridional direction geopotential height is essentially constant. Assume that both Cape Town and Caledon are situated at 34°S. Where relevant, answer the following to the nearest whole number in $\text{m} \cdot \text{s}^{-1}$:
1. What would be the approximate geostrophic wind direction bearing in degrees? [1 mark]
 2. What is the absolute value of the Coriolis parameter $|f|$ at this latitude, to 3 significant digits? [2 marks]
 3. Using the value for $|f|$ given above, estimate the mean geostrophic wind speed between Cape Town and Caledon. [3 marks]

Short answer questions:

- (B) Refer to the Skew-T log-P diagram above. [6 marks].
1. Read off the dew point temperature at the 700hPa level [1 mark].
 2. Where is there most likely to be cloud present in the environmental profile at the time of measurement? Provide a pressure level in hPa. [1 mark]
 3. To what pressure level would a surface air parcel need to be lifted in this environment before it would continue to rise by itself? [1 mark]
 4. If a surface air parcel were lifted adiabatically beyond the height at which saturation is achieved and clouds began to form, what would the pressure level be at the base of these clouds? [1 mark]
 5. Is the atmospheric boundary layer stable, absolutely unstable, conditionally unstable, or neutrally stable with respect to a dry air parcel? [1 mark]
 6. What is the pressure level at the base of the capping inversion layer? [1 mark]

Answer calculation and short-answer questions below:

Calculation answers:

A1: _____

A2: formula used: _____

substitution: _____

answer: _____

A3: formula used: _____

substitution: _____

answer: _____

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Short-answers

B1: _____

B2: _____

B3: _____

B4: _____

B5: _____

B6: _____

For dynamics:

$$\begin{aligned} |U_g| &= \frac{1}{|f|\rho} \left| \frac{\partial P}{\partial n} \right| = \frac{g}{|f|} \left| \frac{\partial Z}{\partial n} \right| & u_g &= -\frac{1}{f\rho} \frac{\partial P}{\partial y} = -\frac{g}{f} \frac{\partial Z}{\partial y} & v_g &= \frac{1}{f\rho} \frac{\partial P}{\partial x} = \frac{g}{f} \frac{\partial Z}{\partial x} \\ |U_t| &= \frac{R}{|f|} \left| \frac{dT}{dn} \right| \left(\ln \left(\frac{P_0}{P} \right) \right) & v_t &= \frac{R}{f} \frac{\partial T}{\partial x} \left(\ln \left(\frac{P_0}{P} \right) \right) & u_t &= -\frac{R}{f} \frac{\partial T}{\partial y} \left(\ln \left(\frac{P_0}{P} \right) \right) \\ \frac{\zeta + f}{D} &= \text{constant} & f &= 2\Omega \sin \phi & \Omega &= \frac{2\pi}{(24)(60)^2} \text{s}^{-1} \\ \text{Div} &= \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) & \zeta &= \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \end{aligned}$$

Rest of the formula sheet on next page

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1 Formula sheet

$$\begin{aligned}
 P &= \rho R T_v & T_v &= T(1 + c_v X); c_v = 0.61 & \frac{\partial P}{\partial z} &= -\rho g \\
 \Delta z &\approx \frac{R}{g} \bar{T}_v \ln\left(\frac{P_0}{P_1}\right) \\
 E_{\text{OLR}} &= \epsilon \sigma T^4 & \sigma &= 5.67 \times 10^{-8} \text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-4} \\
 \lambda_{\text{max}} &= \frac{C}{T} & C &= 2897 \mu\text{m} \cdot \text{K} = 2.897 \times 10^{-3} \text{m} \cdot \text{K} \\
 g &= 9.8 \text{m} \cdot \text{s}^{-2} & R &= 287 \text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1} & S &= 1367 \text{W} \cdot \text{m}^{-2} \\
 P &= \frac{F}{A} = \frac{mg}{A} & \rho &= \frac{m}{V} \\
 F^* &= F_G - F_H - F_E \\
 \Gamma &= -\frac{\partial T}{\partial z} & P_1 &= P_0 \left(\frac{T_1}{T_0}\right)^b & b &= \frac{g}{\Gamma R} \\
 z &= \frac{R_E Z}{R_E - Z} & R_E &= 6.4 \times 10^6 \text{m} \\
 e_s &= e_0 \exp\left(\frac{L_v}{R_v} \left(\frac{1}{T_0} - \frac{1}{T}\right)\right) \\
 T_0 &= 0^\circ\text{C} = 273.15\text{K} & e_0 &= e_s(T = 0^\circ\text{C}) = 611.3\text{Pa} \\
 L_v &= 2.50 \times 10^6 \text{J} \cdot \text{kg}^{-1} & R_v &= 461 \text{J} \cdot \text{K}^{-1} \cdot \text{kg}^{-1} \\
 \text{RH} &= 100 \frac{q}{q_s} = 100 \frac{e}{e_s} \approx 100 \frac{X}{X_s} \\
 q &= \frac{(\eta)(e)}{P - (e)(1 - \eta)} = \frac{X}{1 + X} & \eta &= \frac{M_{\text{H}_2\text{O}}}{M_{\text{air}}} = 0.622 \\
 e &= \frac{(X)(P)}{\eta + X} & X &= \frac{(\eta)(e)}{P - e} \\
 \Theta &= T + \Gamma_d (Z - Z_{1000\text{hPa}}) = T \left(\frac{P_0}{P}\right)^{\frac{\Gamma_d}{C_{p,d}}} & \frac{R}{C_{p,d}} &= 0.286; & P_0 &= 1000\text{hPa} \\
 \text{CAPE} &= g \int_{Z_{\text{LFC}}}^{Z_{\text{EL}}} \frac{T_{v,p} - T_{v,e}}{T_{v,e}} dZ \approx g \sum_{Z_{\text{LFC}}}^{Z_{\text{EL}}} \frac{T_{v,p} - T_{v,e}}{T_{v,e}} \Delta Z & w_{\text{max}} &= \sqrt{2(\text{CAPE})} \\
 \text{CIN} &= g \int_{Z_0}^{Z_{\text{LFC}}} \frac{T_{v,p} - T_{v,e}}{T_{v,e}} dZ \approx g \sum_{Z_0}^{Z_{\text{LFC}}} \frac{T_{v,p} - T_{v,e}}{T_{v,e}} \Delta Z \\
 \Theta_e &= \Theta + \frac{(L_v)(\Theta)(X)}{(C_{p,d})(T)}
 \end{aligned}$$

here: $\exp(x) \equiv e^x$

Unless otherwise stated, assume $\Gamma_d = 0.01 \text{K} \cdot \text{m}^{-1}$ and $\Gamma_s = 0.006 \text{K} \cdot \text{m}^{-1}$